Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 103401
Subject: Electrical and Electronics Measurement
Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) PMMC instruments can be used for measuring
- (i) AC only
- (ii) both AC and DC
- (iii) DC only
- (iv) high frequency signals

(b) Precision of a measuring instrument refers to
- (i) correctness of result
- (ii) smallest measurable quantity
- (iii) consistency of repeated measurements
- (iv) speed of response

(c) A meter reads 125 V and the true value of the voltage is 125.5 V. Find the static error of the instrument.
- (i) 125/0.5 V
- (ii) 0.5 V
- (iii) 125 V
- (iv) 0.5/125 V

(d) Kelvin's double bridge is mainly used for measurement of
- (i) high resistance
- (ii) low resistance
- (iii) medium resistance
- (iv) insulation resistance

(e) Hay's bridge is particularly useful for measuring
- (i) low Q factor
- (ii) zero resistance
- (iii) high Q factor
- (iv) capacitance and dielectric losses

(f) A ballistic galvanometer is used for measurement of
- (i) steady current
- (ii) quantity of charge
- (iii) alternating current
- (iv) voltage

© Apna BEU

(g) Logarithmic decrement is related to
- (i) steady deflection
- (ii) control torque
- (iii) oscillation damping
- (iv) magnetic field strength

(h) One major advantage of potentiometer over voltmeter is
- (i) simpler construction
- (ii) no power drawn from circuit
- (iii) faster response
- (iv) lower cost

(i) Current transformers are normally operated with secondary
- (i) open-circuited
- (ii) loaded with capacitor
- (iii) short-circuited
- (iv) connected to battery

(j) In a CRO, a delay line is used to
- (i) increase gain
- (ii) allow viewing of leading edge of waveform
- (iii) improve focusing
- (iv) reduce noise

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2 (a) Define measurement. Explain accuracy, precision, and significant figures with suitable examples. [7]

(b) Explain the measurement of low resistance using Kelvin's double bridge with a neat diagram. [7]

Q.3 (a) Derive the expression for deflecting torque of a PMMC instrument and list its advantages and limitations. [7]

(b) Describe the construction and working principle of electrodynamometer type wattmeter. Discuss errors and their compensation. [7]

Q.4 (a) Describe the methods used for measurement of earth resistance and their importance. [7]

(b) Explain the measurement of self-inductance using ammeter and voltmeter method. [7]

Q.5 (a) Explain Felici's method for measurement of mutual inductance. [7]

(b) Describe the construction and working principle of a D'Arsonval galvanometer. [7]

Q.6 (a) Discuss the effect of resistance on damping in a galvanometer. Explain logarithmic decrement. [7]

(b) A highly sensitive galvanometer can detect a current as low as 0.1 nA. This galvanometer is used in a Wheatstone bridge as a detector. Each arm of the bridge has a resistance of 1 kΩ. The input voltage applied to the bridge is 20 V. Calculate the smallest change in resistance which can be detected. The resistance of the galvanometer can be neglected as compared with the internal resistance of bridge. [7]

Q.7 (a) Explain the construction, working principle, and advantages of DC potentiometers. Describe any one type in detail. [7]

(b) A current transformer has a single turn primary and a 200 turns secondary winding. The secondary supplies a current of 5 A to a non-inductive burden of 12 resistances The requisite flux is set up in the core by an mmf of 80 A. The frequency is 50 Hz and the net cross section of the core is 1000 mm². [7]

Q.8 (a) Describe the construction and operation of a Potential Transformer (PT). How does it differ from a CT? [7]

(b) Explain the working principle of AC potentiometers. Compare DC and AC potentiometers. [7]

Q.9 Write short note on ***any two*** of the following: **[7 x 2 = 14]**

(a) Anderson Bridge
(b) Megohmmeter
(c) Moving iron instrument
(d) Analog and Digital Storage Oscilloscopes

© Apna BEU

HIGHLY CONFIDENTIAL DOCUMENT - 151 - Government Engineering College, Siwan @ 202[illegible]

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play